Clinical Neurophysiology 129 (2018) 2315–2324

Contents lists available at ScienceDirect

Clinical Neurophysiology

journal homepage: www.elsevier.com/locate/clinph

EEG coherence in a mental arithmetic task performance in first episode schizophrenia and schizoaffective disorder

Yuliya Zaytseva [a,b,c,1,*], Zhanna Garakh [d,1], Vladimir Novototsky-Vlasov [d], Isaac Ya. Gurovich [e], Alexander Shmukler [e], Anna Papaefstathiou [b], Jiří Horáček [a,b], Filip Španiel [a,b], Valeria B. Strelets [d]

[a] National Institute of Mental Health, Klecany, Czech Republic
[b] Department of Psychiatry and Psychotherapy, 3rd Faculty of Medicine, Charles University in Prague, Prague, Czech Republic
[c] Human Science Centre, Institute of Medical Psychology, Ludwig-Maximilian University, Munich, Germany
[d] Institute of Higher Nervous Activity and Neurophysiology of Russian Academy of Science, Moscow, Russian Federation
[e] Moscow Research Institute of Psychiatry – Branch of V Serbsky National Medical Research Centre for Psychiatry and Narcology, Moscow, Russian Federation

ARTICLE INFO

Article history:
Accepted 31 August 2018
Available online 6 September 2018

Keywords:
EEG coherence
First psychotic episode
Schizophrenia
Schizoaffective disorder
Resting state
Mental arithmetic task

HIGHLIGHTS

- Neural cognitive processing differs in schizophrenia and schizoaffective disorder.
- Schizophrenia is associated with decreased *alpha* coherence during mental arithmetic task.
- Schizoaffective group showed higher *alpha*, *beta1*, *beta2* and *gamma* coherence.

ABSTRACT

Objectives: To evaluate the coherence values of the cortical bands in patients with first episode schizophrenia and schizoaffective disorder during the performance of a mental arithmetic task.
Method: We analysed EEG coherence in the resting state and subsequently while counting down from 200 in steps of 7 in 32 first episode schizophrenia patients (SZ), 32 patients with first episode schizoaffective disorder (SA) and 40 healthy controls (HC).
Results: Task performance in HC is characterised by decreased coherence in the *alpha* bands and increased coherence in the *beta 2* and *gamma* bands in frontal sites. In SZ, coherence decreases in the *alpha* band, whereas in SA it substantially increases in the *alpha, beta1, beta 2 and gamma* bands.
Conclusions: Despite no differences in performance on a behavioural level, SA patients demonstrate a paradoxical increase in both low and fast frequency bands during the performance of a mental arithmetic task, while, patients with SZ show a decreased coherence in the alpha band, presumably alluding to the excessive excitatory (in SA) and inhibitory (in SZ) mechanisms in cognitive processing.
Significance: The current study provides evidence for the distinction of neurophysiological mechanisms of cognitive processing between SZ and SA.
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1. Introduction

Current theories of schizophrenia (SZ) have emphasized that the pathophysiology arises from a 'dysconnectivity' between and within cortical areas of the brain (Friston, 1999, Uhlhaas and Singer, 2010, Stephan et al., 2009). Studies of functional connectivity in patients with schizophrenia using PET and functional MRI data both during cognitive tasks (Potkin and Ford, 2009 for a review) and at rest (for a review see Greicius, 2008) vastly support this hypothesis. However, less attention has been paid to the particular disorders within the schizophrenia spectrum. To date, only two studies have demonstrated the alterations in the brain connectivity in schizoaffective disorder (SA). In the first study, the authors applied dynamic functional connectivity (DFC) in resting-state fMRI, the connectivity in SA differed from all other groups (healthy controls, schizophrenia and bipolar patients) and specifically in the connections between the frontal and fusiform cortices (Du et al., 2017). The results were supported by another study that showed reduced grey matter volumes of the fusiform gyrus in SA (Landin-Romero et al., 2017).

* Corresponding author at: National Institute of Mental Health, Topolova 748, 250 67 Klecany, Czech Republic.
E-mail address: yuliya.zaytseva@gmail.com (Y. Zaytseva).
[1] These authors contributed equally.

https://doi.org/10.1016/j.clinph.2018.08.020
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The electroencephalographic (EEG) approach to connectivity is based on coherence and phase locking analysis between EEG electrode sites, it has become a reliable way of detecting abnormalities in schizophrenia (Bowyer, 2016). Coherence reflects the coordinal inter-relation between brain networks and is closely linked to cognitive processing, as EEG/MEG parameters allow the quantification of interactions between sensor/source-estimates of rhythmic activity (Decker et al., 2017; Fries, 2005, 2015). Coherent neuronal connections capitalize on the neurotransmission dynamics driven by amino acids glutamate, GABA and others (Maran et al., 2016). This method was successfully applied for delineation of schizophrenia and bipolar disorder in resting state EEG (Kam et al., 2013). Specifically, Kam et al. (2013), demonstrated an increased power in high frequency bands with few disruptions in neural synchronization in bipolar patients. In contrast, schizophrenia patients generally exhibited enhanced synchronization of high frequency bands within and across hemispheres. The alterations in the resting state coherence in various frequency bands, measured by EEG/MEG in patients with schizophrenia, has been intensively studied, although it remains contradictory (Boutros et al., 2008; Winterer et al., 2001). The studies suggest that alterations in fast oscillations at beta and gamma band frequencies are elevated at rest (Di Lorenzo et al., 2015), reduced (Kam et al., 2013) or even intact (Winterer et al., 2001).

It is well established that cognitive dysfunction constitutes a hallmark feature of schizophrenia and schizophrenia spectrum disorders (Heinrichs and Zakzanis, 1998; Keefe, 2008). Neurocognitive deficits in schizophrenia patients correspond to structural and functional changes in the brain (van Haren et al., 2008; Lewis, 2012; Schultz et al., 2012). Analogously to schizophrenia, schizoaffective disorder (SA) has been associated with cognitive decline in the same cognitive domains, although it is less pronounced than in schizophrenia (for a review see Madre et al., 2016).

The neurophysiological mechanisms underlying cognitive dysfunctions and their distinct profiles in schizophrenia and schizophrenia spectrum disorders are not well understood (Garakh et al., 2015; Maran et al., 2016). In cognitive tasks, patients with schizophrenia exhibited altered EEG coherence during the processing of acoustic stimuli or recognition of emotional expression (Brenner et al., 2009, Popov et al., 2014). A task that would probe executive functioning or working memory (DeStefano and LeFevre, 2004), presumably being the mostly affected in schizophrenia and schizophrenia spectrum disorders, has not been widely applied. In one early report, Morrison-Stewart and authors used a mental arithmetic task and suggested an aberrant functional organization of the brain in schizophrenia, particularly affecting the left hemisphere (Morrison-Stewart et al., 1991). Although a mental arithmetic task is easy to administer, it is a complex task that involves attention, working memory, and executive functions, which are necessary for performance of even simple mathematical operations such as addition or subtraction. Therefore, its application may be beneficial to test the hypothesis of dysconnectivity in schizophrenia and schizophrenia spectrum disorders.

The utility of a task-related EEG coherence approach in understanding the variability of cognitive processing in clinical populations has not yet been applied; however, we assume that functional connectivity changes during task processing presumably having various trajectories (Shmukler et al., 2015) may contribute to the understanding of the common and the distinct mechanisms of cognitive dysfunctions in the schizophrenia spectrum disorders and serve for differentiation of the disorders. The aim of the study was to evaluate the variation of coherence during a task-induced activity and resting condition in clinically homogenous groups of patients with first episode schizophrenia and schizoaffective disorder. We hypothesized that the groups may differ in cognitive task processing. In addition, inclusion of patients with first episode psychosis allowed us to study EEG correlates regardless of the confounding effect of medication and duration of the illness.

2. Methods

2.1. Experimental sample

The study was conducted in a sample of patients with first episode psychosis and healthy volunteers as previously described in Garakh et al (2015). Patients in their first psychotic episode diagnosed with schizophrenia (n = 32) and schizoaffective disorder of both manic and depressive episodes (n = 32) were enrolled into the study. The psychopathology was evaluated in the routine clinical examination and Structural Clinical Interview for PANSS. Diagnosis was based on ICD-10 and was confirmed in a 1-year follow-up.

Control group consisted of 40 healthy volunteers without diagnosis of mental illness or neurological disorder or brain injuries. The groups of patients and controls were matched by gender, age and years of education. Demographic and clinical characteristics of the samples are given in Table 1. The study was approved by the local ethics committee and all of the participants read and signed the informed consent.

2.2. EEG acquisition, experimental task and data analysis

Electroencephalogram was recorded from 19 sites based on a standard 'International 10–20 System' (Jasper, 1958): Fp1, Fp2, F3, F4, F7, F8, C3, C4, T3, T4, T5, T6, P3, P4, O1, O2, including three midline sites (Fz, Cz, Pz). Reference electrodes were placed on the linked earlobes. Low pass filters set at 70 Hz with a time constant of 0.3 s were used during the recording. EEG mapper (MBM, Russia) was used to quantify EEG traces of 100 s. with a 200 Hz sampling rate. The electrode impedances were equal or below 10 kΩ, with an average of 5 kΩ. For four patients and four controls the data were obtained from 16 cortical electrodes with the same technical characteristics. Post-processing was performed using a custom designed variant of the multi-source eye correction method (Novototsky-Vlasov et al., 2007) with a subsequent rejection of the EEG segments with artefacts (eye movements, muscle artefacts). Finally, 10–15 five-second artefact-free sets (epochs/intervals) were selected by an expert for further analysis.

EEG was recorded during the resting state and during the task performance. In the resting state the subjects were instructed to close their eyes and not think of anything in particular. In the experimental task the participants were asked to perform a mental arithmetic task known as Kraepelinean counting, in which they had to perform a subsequent subtraction from 200 in steps of 7 during a 100-sec. time interval. After the task the subjects underwent a debriefing, in which they were asked for the actual result of the count. The detailed description of the task is given in Garakh et al (2015).

Custom designed software was used for coherence computation. The data were analysed using coherence analysis of 6 frequency bands: *delta* (1.5–3.9 Hz), *theta* (4–7 Hz), *alpha* (8–13 Hz), *beta1* (13–20 Hz), *beta2* (20–30 Hz) and *gamma* (30–40 Hz). Coherence was calculated using standard algorithms (Bendat and Piesol, 2010):

$$Coh(c1,c2)(f) = |CS_{c1,c2}(f)|^2/(CS_{c1,c1}(f)^*|CS_{c2,c2}(f)),$$
$$where\ \ CS_{c1,c2}(f) = \sum c1,\ \ i(f)c2,\ \ i(f),$$

where CS – complex spectrum; c1 (c2) – channel 1 (2); f – frequency; i – epoch number.

Table 1
Demographic and clinical parameters of the sample.

Variables	Groups			P	
	Schizophrenia (SZ) *n* = 32	Schizoaffective disorder (SA) *n* = 32	Healthy control (HC) subjects *n* = 40		
Age (years), mean ± SD	28.91 ± 10.64	27.59 ± 6.93	27.63 ± 6.37	0.752[a]	HC = SZ; HC = SA; SZ = SA
Education (years), mean ± SD	13.47 ± 1.50	13.75 ± 1.87	14.05 ± 6.37	0.437[a]	HC = SZ; HC = SA; SZ = SA
Handedness	32 right	32 right	40 right	1.00	HC = SZ; HC = SA; SZ = SA
Gender, males (females)	16(16)	16(16)	20(20)	1.00	HC = SZ; HC = SA; SZ = SA
PANSS positive	19.34 ± 5.58	15.34 ± 4.55		0.002[b]	SZ > SA
PANSS negative	17.09 ± 4.62	13.97 ± 4.48		0.007[b]	SZ > SA
PANSS general	41.50 ± 7.55	37.41 ± 6.71		0.025[b]	SZ > SA
Diagnosis subtype (ICD-10)	Paranoid schizophrenia (n = 32)	manic type (n = 5), depressive type (n = 27)			
Age at illness onset, mean ± SD	28.78 ± 10.54	26.65 ± 6.77		0.299[b]	SZ = SA
Chlorpromazine equivalent (CPZ), mean ± SD	227.55 ± 69.10	208.62 ± 80.90		0.361[b]	SZ = SA

Note: SD – standard deviation.
[a] Analysis of variance (one-way ANOVA by three groups).
[b] t test independent by group.

For the same frequency bands, coherence values were computed from the cross spectrum of selected electrode pairs. The coherence between all site combinations was computed in 171 pairs, then all of the EEG coherence values were transformed using an Arctan transform in order to provide the normal distribution. The interhemispheric coherence was analysed in total of 64 pairs of sites i.e. in 28 pairs of sites of the left and 28 pairs of sites of the right hemisphere as well as between and within the midline sites (Fz, Cz, Pz).

2.3. Statistical analysis

Statistical analysis was performed using STATISTICA 10.0 software. Demographic and clinical parameters (PANSS scores) were compared between the groups using a t-test. Separate repeated measures ANOVAs were conducted for the inter-hemispheric, intra-hemispheric and midline site pairs for each of the six frequency bands. To examine the inter-hemispheric and midline coherence, we conducted ANOVAs including one between-subject factor (group) and two within-subject factors (condition: rest, task; pairs of sites). To examine the intra-hemispheric coherence, we conducted ANOVAs including one between-subject factor (group) and three within-subject factors (condition: rest, task; electrode pairs; hemisphere: left, right). To follow up significant interactions, we conducted one-way or two-way ANOVAs for each condition or group. Greenhouse-Geisser corrections were applied to correct for violations of sphericity and homogeneity. Repeated Measures ANOVAs was performed with FDR correction and was significant at the level $p \leq 0.001$ and $p < 0.0021$ for between (48 comparisons) and within group (24 comparisons) comparisons, respectively. Significant main effects were followed up with post-hoc between-group and within-group comparisons, which were adjusted using a Bonferroni correction.

Explorative correlations were obtained with the sum of positive, negative and total score PANSS and the coherence values of the patients' groups that deviated from the coherence values of the healthy individuals. Correlations between EEG coherence values in the resting state and the in-task condition and PANSS scores were obtained for each diagnostic group using Pearson correlation coefficients with subsequent FDR correction.

3. Results

Behavioural task performance results have previously been thoroughly reported by our group (Garakh et al., 2015). Briefly, based on the debriefing, the number of the correct answers did not differ between the groups ($p = 0.82$), although the speed of counting was significantly higher in the controls ($p \leq 0.05$).

Repeated measures ANOVAs revealed the main effect of 'GROUP' and the main effect of 'CONDITION' and their interaction in the *alpha, beta1, beta2 and gamma* frequency bands for inter-hemispheric, intra-hemispheric and midline coherence values. The results of the ANOVAs are summarized in Table 2.

3.1. Between-group comparisons of coherence in the resting state

Significant differences of inter-hemispheric coherence between the three groups were found in the *alpha, beta1, beta2 and gamma* bands and for intra-hemispheric and midline coherence in the *beta1, beta2 and gamma* bands. Follow-up tests results are summarized in Table 3.

HC vs. SZ comparison. Interhemispheric coherence was higher in HC in the *alpha beta*1 *beta*2 and *gamma* bands in the anterior sites, with the exception of the connections between P3-P4 in the *beta2* band, and C3-P4, P3-P4 in the *gamma* band, which was higher in SZ. In terms of intra-hemispheric coherence, differences were determined in the *beta2* and *gamma* bands, which were higher in the anterior sites in HC than in SZ. *Beta1* band intra-hemispheric coherence was higher in HC in the anterior sites and lower in the P3-T5 sites. For the midline sites, coherence was higher in HC in the *beta1* band. In the *beta2* band, the midline coherence in HC was higher in the Fp2-Fz and lower in the P4-Pz sites. For the *gamma* band differences were found in the connections between the particular sites, notably C4-Pz, P3-Pz, P4-Pz, being lower in HC.

HC vs. SA comparison. Differences in inter-hemispheric coherence in the anterior sites were found in the *beta1* and *beta2* bands, with the values being higher in HC than in SA patients. In the *gamma* band, interhemispheric coherence was lower in P3-P4 in HC. *Alpha* band interhemispheric coherence was not found to be significant. Coherence of the intra-hemispheric connections differed significantly in the *beta1*, beta2 and *gamma* bands. Coherence

Table 2
Repeated measures ANOVA of inter-hemispheric, intra-hemispheric and midline coherence among SZ, SA and HC.

	Frequency band					
Source	Delta	Theta	Alpha	Beta1	Beta2	Gamma
Inter-hemispheric						
Group	F(2,101) = 0.865 p = 0.420	F(2,101) = 1.870 p = 0.159	F(2,101) = 9.406 **p < 0.001**	F(2,101) = 5.569 **p = 0.005**	F(2,101) = 1.790 p = 0.172	F(2,101) = 0.278 p = 0.758
Condition × group	F(2,101) = 0.062 p = 0.940	F(2,101) = 1.004 p = 0.370	F(2,101) = 4.347 **p = 0.015**	F(2,101) = 4.061 **p = 0.020**	F(2,101) = 2.114 p = 0.126	F(2,101) = 2.635 p = 0.077
Condition × pair × group	F(126,6363) = 1.261 p = 0.124	F(126,6363) = 0.953 p = 0.573	F(126,6363) = 0.937 p = 0.535	F(126,6363) = 1.260 p = 0.224	F(126,6363) = 1.662 **p = 0.047**	F(126,6363) = 1.858 **p = 0.015**
Intra-hemispheric						
Group	F(2,101) = 0.069 p = 0.994	F(2,101) = 1.583 p = 0.210	F(2,101) = 3.779 **p = 0.026**	F(2,101) = 4.848 **p = 0.009**	F(2,101) = 2.731 p = 0.070	F(2,101) = 1.905 p = 0.154
Condition × group	F(2,101) = 0.760 p = 0.470	F(2,101) = 1.079 p = 0.344	F(2,101) = 4.112 **p = 0.019**	F(2,101) = 4.313 **p = 0.016**	F(2,101) = 0.736 p = 0.482	F(2,101) = 1.815 p = 0.168
Condition × hemisphere × group	F(2,101) = 0.061 p = 0.941	F(2,101) = 0.047 p = 0.955	F(2,101) = 0.109 p = 0.897	F(2,101) = 0.869 p = 0.423	F(2,101) = 0.511 p = 0.602	F(2,101) = 0.477 p = 0.622
Condition × pair × group	F(54,2727) = 0.865 p = 0.666	F(54,2727) = 1.067 p = 0.366	F(54,2727) = 0.384 p = 0.972	F(54,2727) = 1.409 p = 0.144	F(54,2727) = 2.098 **p = 0.012**	F(54,2727) = 2.544 **p < 0.001**
Condition × hemisphere × pair × group	F(54,2727) = 1.096 p = 0.334	F(54,2727) = 1.014 p = 0.447	F(54,2727) = 0.328 p = 0.983	F(54,2727) = 0.462 p = 0.946	F(54,2727) = 0.752 p = 0.740	F(54,2727) = 1.033 p = 0.418
Midline						
Group	F(2,87) = 1.226 p = 0.299	F(2,87) = 2.574 p = 0.082	F(2,87) = 2.939 p = 0.058	F(2,87) = 3.919 **p = 0.023**	F(2,87) = 0.473 p = 0.624	F(2,87) = 0.831 p = 0.439
Condition × group	F(2,87) = 1.798 p = 0.172	F(2,87) = 2.583 p = 0.081	F(2,87) = 3.398 **p = 0.038**	F(2,87) = 6.998 **p = 0.002**	F(2,87) = 3.368 **p = 0.039**	F(2,87) = 3.766 **p = 0.027**
Condition × pair × group	F(100,4350) = 0.838 p = 0.707	F(100,4350) = 0.462 p = 0.961	F(100,4350) = 0.722 p = 0.796	F(100,4350) = 1.182 p = 0.255	F(100,4350) = 1.498 p = 0.069	F(100,4350) = 1.619 **p = 0.043**

Table 3
Repeated measures ANOVA of inter-hemispheric, intra-hemispheric and midline coherence in Rest Condition among SZ, SA and HC.

	Frequency band			
Source	Alpha	Beta1	Beta2	Gamma
Inter-hemispheric				
Group	F(2,101) = 8.580 **p < 0.001**	F(2,101) = 7.988 **p < 0.001**	F(2,101) = 1.586 p = 0.210	F(2,101) = 0.077 p = 0.926
Pair × group	F(126, 6363) = 1.834 p = 0.036	F(126, 6363) = 5.705 **p < 0.001**	F(126, 6363) = 4.206 **p < 0.001**	F(126, 6363) = 2.837 **p < 0.001**
Intra-hemispheric				
Group	F(2,101) = 3.043 p = 0.052	F(2,101) = 3.322 p = 0.040	F(2,101) = 2.551 p = 0.083	F(2,101) = 1.171 p = 0.314
Hemisphere × group	F(2,101) = 2.310 p = 0.105	F(2,101) = 0.988 p = 0.376	F(2,101) = 0.436 p = 0.648	F(2,101) = 0.302 p = 0.740
Pair × group	F(54,2727) = 1.702 p = 0.056	F(54,2727) = 4.419 **p < 0.001**	F(54,2727) = 3.534 **p < 0.001**	F(54,2727) = 3.016 **p < 0.001**
Hemisphere × pair × group	F(54,2727) = 0.685 p = 0.771	F(54,2727) = 0.850 p = 0.619	F(54,2727) = 1.097 p = 0.355	F(54,2727) = 1.044 p = 0.407
Midline				
Group	F(2,87) = 2.667 p = 0.075	F(2,87) = 3.010 p = 0.054	F(2,87) = 0.026 p = 0.974	F(2,87) = 1.394 p = 0.254
Pair × group	F(100,4350) = 1.105 p = 0.346	F(100,4350) = 3.493 **p < 0.001**	F(100,4350) = 2.503 **p < 0.001**	F(100,4350) = 2.553 **p = 0.001**

of the *beta1* and *beta2* bands in the anterior sites was higher in HC, while coherence in the local connections of the *beta1* band in P3-T5, T3-T5 and the *beta2* band in P3-T5, P4-T6 was lower in HC. No significant differences between HC and SA in the coherence of the midline sites.

SZ vs. SA comparison. Differences were determined in inter-hemispheric coherence and midline-coherence in the *beta1* band, being lower in SZ compared to SA. Intra-hemispheric coherence was lower in SZ in the particular connections C3-T3, P3-O1, F4-C4, C4-T4 of the *beta1* band P3-O1 of the *gamma* band.

A summary of the post-hoc analysis of the inter-hemispheric, intra-hemispheric and midline coherence values is illustrated in Fig. 1. In all the figures included in this study, error probabilities are mapped onto schematic brain maps as connecting lines between the electrodes involved. Increased coherence is indicated by a brown line ($p < 0.01$) and a red line ($p < 0.05$), while decreased coherence is represented by a violet line ($p < 0.01$) and a blue line ($p < 0.05$).

3.2. Coherence during the arithmetical task performance

The between-group comparison revealed inter-hemispheric, intra-hemispheric and midline coherence differences in the *alpha, beta1, beta2 and gamma* bands. The results of post-hoc tests are given in Table 4.

HC vs. SZ comparison. The inter-hemispheric coherence was higher in HC in anterior sites of the *alpha* band as well as the anterior sites of *beta1* band and *beta2* band. In the *gamma* band the differences were found in the anterior and posterior sites: in the anterior sites the inter-hemispheric coherence was higher in HC, while in the P3-P4 sites it was higher in SZ. Differences in intra-hemispheric coherence were pronounced in the *alpha* band, being

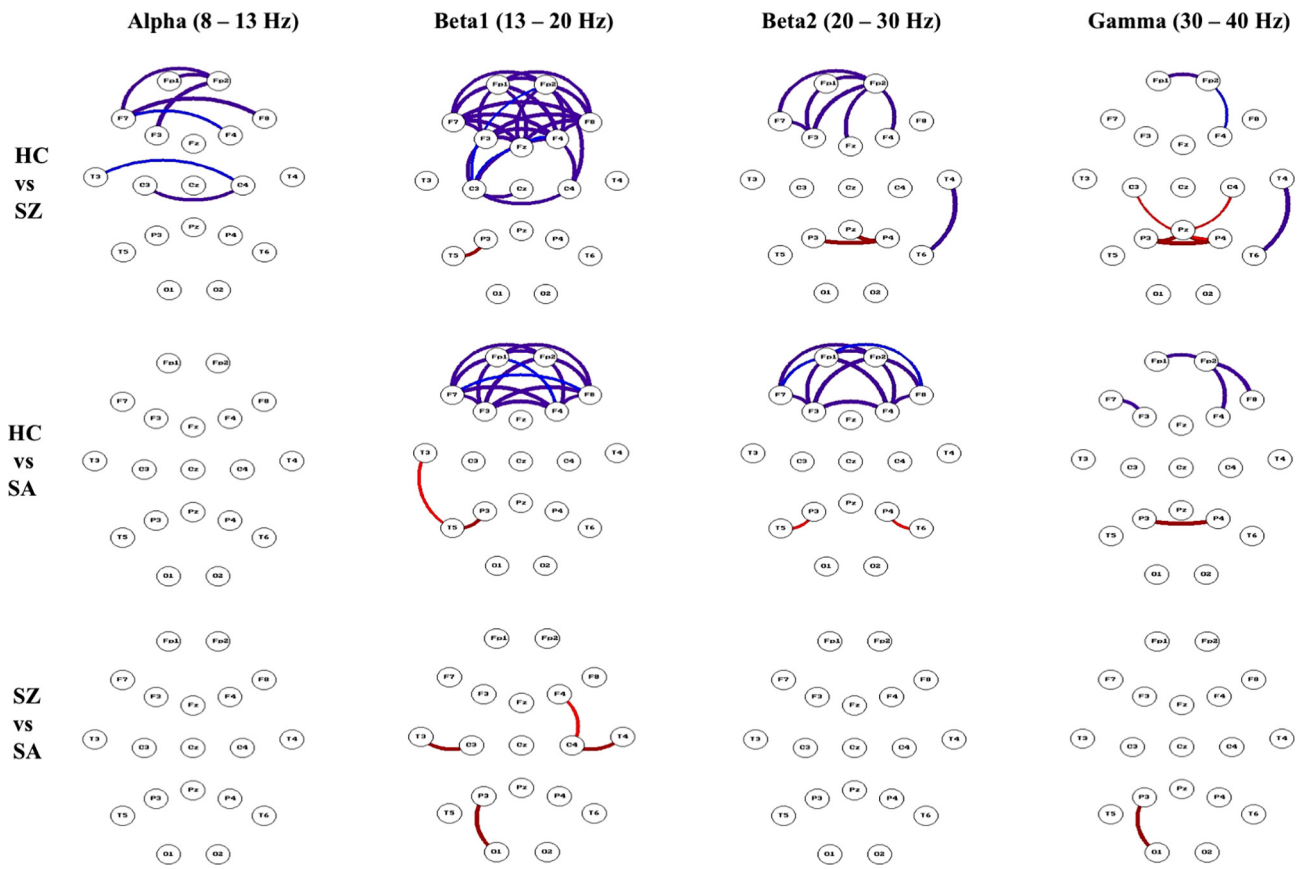

Fig. 1. Between- group comparison of the coherence values in the resting state. Increased coherence – brown line ($p < 0.01$), red line ($p < 0.05$). Decreased coherence – violet line ($p < 0.01$), blue line ($p < 0.05$). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 4
Repeated measures ANOVA of inter-hemispheric, intra-hemispheric and midline coherence in Task Condition among SZ, SA and HC.

	Frequency band			
Source	Alpha	Beta1	Beta2	Gamma
Inter-hemispheric				
Group	F(2,101) = 9.013 **p < 0.001**	F(2,101) = 3.417 p = 0.037	F(2,101) = 1.986 p = 0.143	F(2,101) = 1.149 p = 0.321
Pair × group	F(126, 6363) = 2.396 **p = 0.004**	F(126, 6363) = 5.947 **p < 0.001**	F(126, 6363) = 3.302 **p < 0.001**	F(126, 6363) = 2.190 **p = 0.004**
Intra-hemispheric				
Group	F(2,101) = 4.435 **p = 0.014**	F(2,101) = 6.052 **p = 0.003**	F(2,101) = 2.492 p = 0.088	F(2,101) = 2.429 p = 0.093
Hemisphere × group	F(2,101) = 2.064 p = 0.132	F(2,101) = 1.453 p = 0.239	F(2,101) = 1.349 p = 0.264	F(2,101) = 1.415 p = 0.248
Pair × group	F(54,2727) = 1.816 p = 0.037	F(54,2727) = 4.358 **p < 0.001**	F(54,2727) = 2.595 **p = 0.001**	F(54,2727) = 1.822 **p = 0.029**
Hemisphere × Pair × group	F(54,2727) = 0.978 p = 0.477	F(54,2727) = 1.025 p = 0.427	F(54,2727) = 1.184 p = 0.282	F(54,2727) = 1.146 p = 0.312
Midline				
Group	F(2,87) = 3.215 p = 0.045	F(2,87) = 5.072 **p = 0.008**	F(2,87) = 1.422 p = 0.247	F(2,87) = 1.052 p = 0.354
Pair × group	F(100,4350) = 1.457 p = 0.113	F(100,4350) = 4.443 **p < 0.001**	F(100,4350) = 2.824 **p < 0.001**	F(100,4350) = 2.426 **p = 0.001**

higher in HC. In other bands, *beta1*, *beta2* and *gamma*, the coherence was decreased with the exception of P3-T5, which was higher in these bands in SZ. In the midline sites, the coherence was higher in HC in the *alpha* and *beta1* bands. In the *beta2* and *gamma* bands, coherence was higher in the majority of sites in HC, except for P4-Pz in the *beta2* and P3-Pz, P4-Pz, T5-Pz in the *gamma* band, which were higher in SZ.

HC vs. SA comparison. Heathy controls had a higher inter-hemispheric coherence in the *beta2* band for Fp1-Fp2 sites. Intra-hemispheric coherence was higher in HC in Fp2-F8, F4-F8 and lower in the P3-T5, T3-T5 sites of the *beta1* band, although intra-hemispheric coherence in HC was higher in the *beta2* band in the F4-F8 sites. The midline coherence was no significantly different in the groups.

SZ vs. SA comparison. In SZ patients, the inter-hemispheric coherence was lower than in SA in the *alpha* and *beta1* bands. In the same manner, the intra-hemispheric and midline coherence was lower in SZ in *beta1* bands.

A summary of the post-hoc analysis of inter-hemispheric, intra-hemispheric and midline coherence values during task performance is included in Fig. 2.

3.3. Within-group comparison of coherence changes under task vs. resting conditions

During the task performance the healthy controls (HC) exhibited a decrease of intra-hemispheric [CONDITION × PAIR, $F(27,1053) = 5.31$, $p < 0.0001$] and midline coherence [CONDITION × PAIR, $F(50,1750) = 2.811$, $p = 0.005$] in the *alpha* band. At the same time, the midline coherence was increased in *beta2* [CONDITION × PAIR, $F(50,1750) = 3.080$, $p = 0.002$] and *gamma* [CONDITION × PAIR, $F(50,1750) = 2.393$, $p = 0.013$] bands.

The results were more diverse in the patient groups. In the SZ group, there was only a substantial decrease of *alpha* band inter-hemispheric coherence [CONDITION × PAIR, $F(63,1953) = 3.629$, $p = 0.002$], intra-hemispheric coherence [CONDITION × PAIR, $F(27,873) = 6.005$, $p < 0.001$], midline coherence [CONDITION × PAIR, $F(50,1350) = 4.351$, $p < 0.001$]. On the contrary, during the task performance, SA patients exhibited an increase in inter-hemispheric [CONDITION × PAIR, $F(63,1953) = 3.468$, $p = 0.008$] and midline [CONDITION × PAIR, $F(50,1250) = 3.690$, $p = 0.001$] coherence in the *alpha* band; inter-hemispheric [CONDITION, $F(1,31) = 8.259$, $p = 0.007$], intra-hemispheric [CONDITION, $F(1,31) = 5.520$, $p = 0.025$], and midline coherence [CONDITION, $F(1,25) = 19.410$, $p < 0.001$] in the *beta1* band; inter-hemispheric [CONDITION , $F(1,31) = 7.430$, $p = 0.011$; CONDITION × PAIR, $F(63,1953) = 3.478$, $p = 0.002$], intra-hemispheric [CONDITION × PAIR, $F(27,837) = 4.757$, $p < 0.001$], and midline coherence [CONDITION, $F(1,25) = 10.209$, $p = 0.004$; CONDITION × PAIR, $F(50,1250) = 3.768$, $p = 0.001$] in the *beta2* band, and inter-hemispheric [CONDITION, $F(1,31) = 5.497$, $p = 0.026$; CONDITION × PAIR, $F(63,1953) = 3.567$, $p < 0.001$], intra-hemispheric [CONDITION × PAIR, $F(27,837) = 4.577$, $p < 0.001$], and midline [CONDITION, $F(1,25) = 6.058$, $p = 0.021$; CONDITION × PAIR, $F(50,1250) = 2.743$, $p = 0.008$] in the *gamma* band. A summary of the results is provided in Fig. 3.

3.4. Correlation analysis between total PANSS scores and EEG coherence parameters under resting and task conditions

Given the obtained differences in the coherence values between patients' groups and healthy controls, we analysed the correlations between 134 coherence values for inter-hemispheric, intra-hemispheric and midline connections and PANSS scores values (sum of positive, sum of negative and total PANSS score) in SZ group and 42 values and above mentioned PANSS parameters in SA. FDR correction for SZ group was $p = 0.00015$ and for SA – $p = 0.0004$. The significant negative correlations were obtained only in SZ group and only in the task condition in inter-hemispheric connections in the *beta1* band and positive symptoms score (F4-F7, $r = -0.638$, $p < 0.05$; F7-F8, $r = -0.597$, $p < 0.05$) (Fig. 4).

4. Discussion

This is the first study to examine the differences between coherence in first episode schizophrenia and schizoaffective disorder

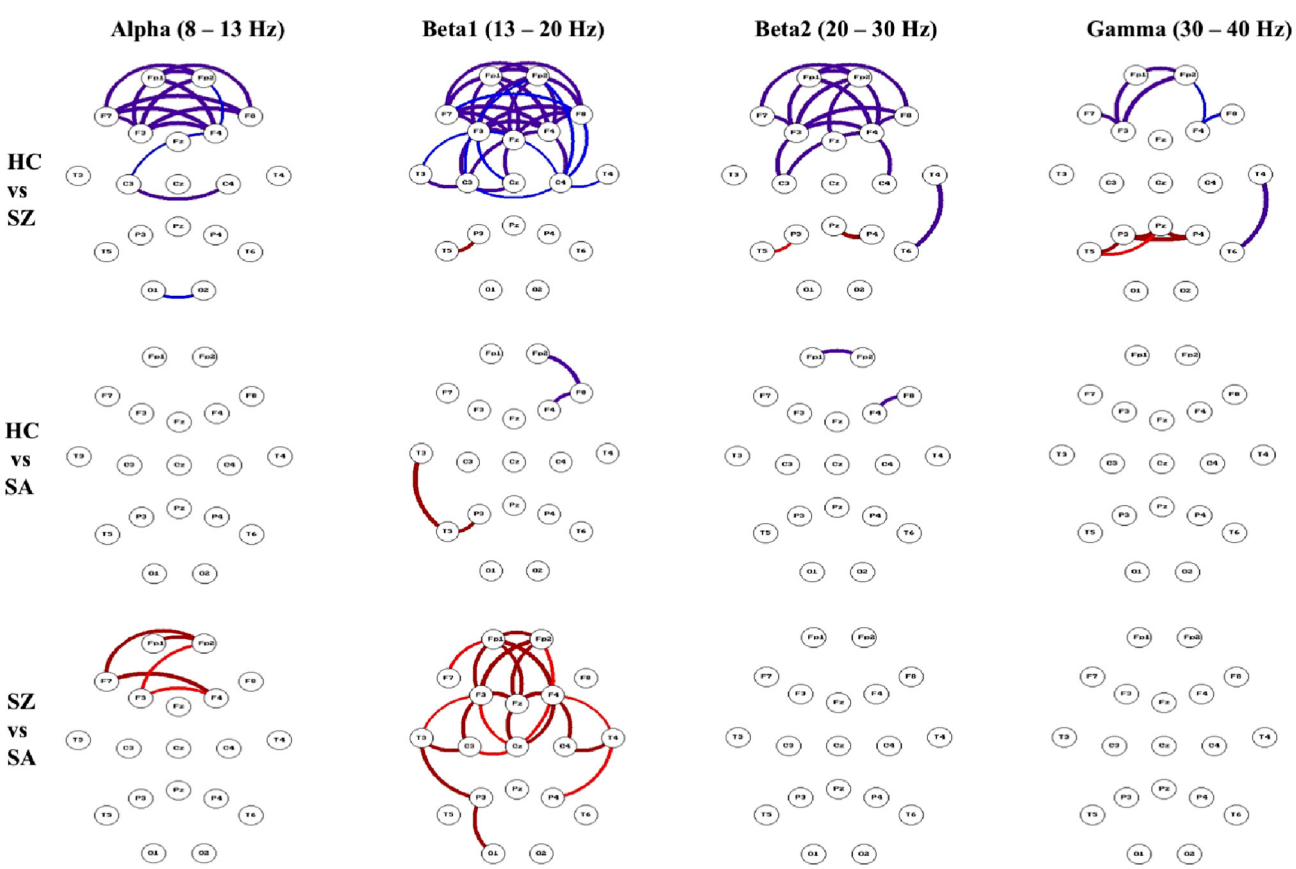

Fig. 2. Between- group comparison of the coherence values in task condition. Increased coherence – brown line ($p < 0.01$), red line ($p < 0.05$). Decreased coherence – violet line ($p < 0.01$), blue line ($p < 0.05$). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

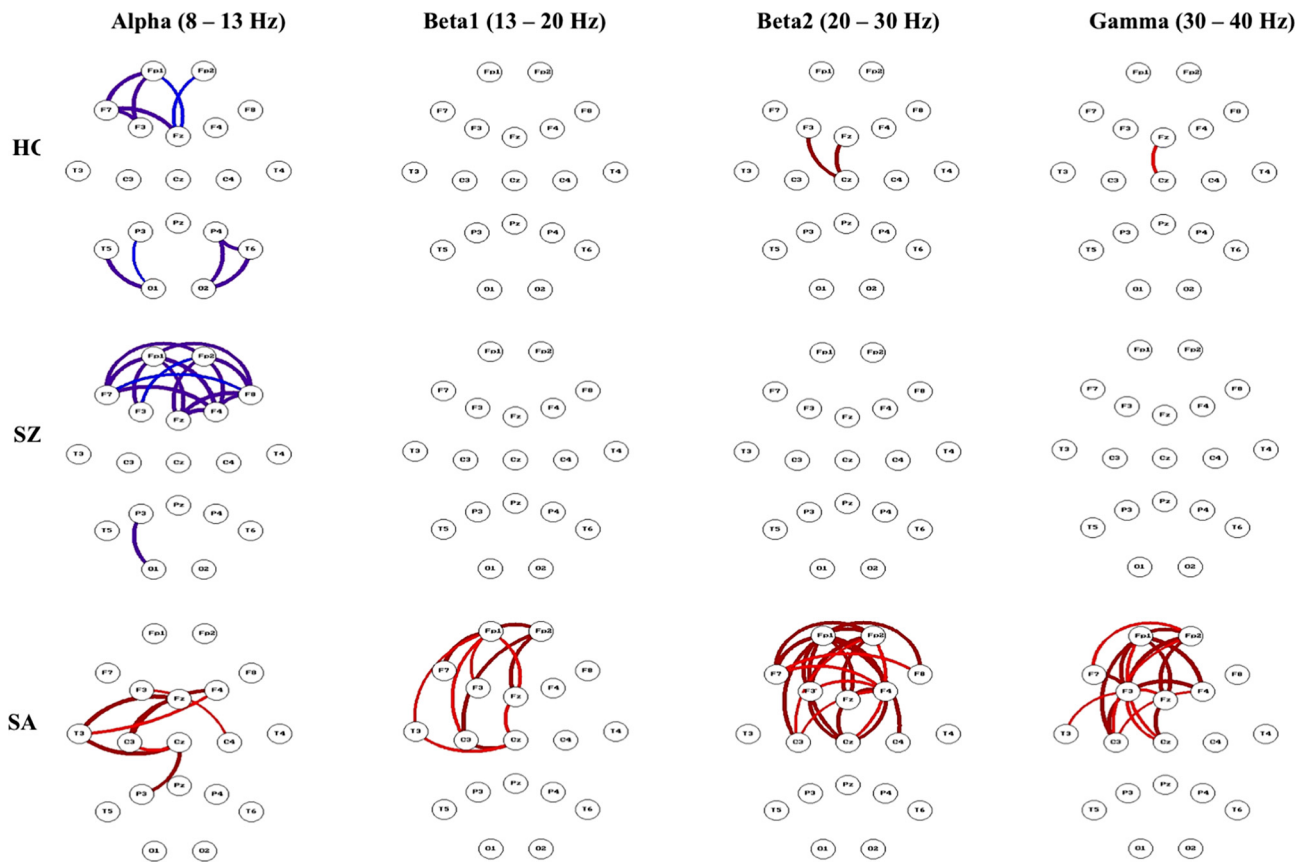

Fig. 3. Changes of the coherence in task vs rest conditions. Increased coherence – brown line ($p < 0.01$), red line ($p < 0.05$). Decreased coherence – violet line ($p < 0.01$), blue line ($p < 0.05$). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

during both the resting state and the performance of an arithmetic cognitive task. Our results provide support for neurophysiological distinctions between both psychotic disorders.

In the resting state, SZ patients presented lower coherence in the *alpha, beta-1, beta-2* bands compared to HC. SA patients showed similar patterns to SZ patients in these bands. Comparison of SZ and SA patients showed increased coherence in SA as opposed to SZ in high frequency (*beta-gamma*) bands in posterior regions. Altered resting state coherence in schizophrenia is well documented in the literature (Hillebrand et al., 2012, Douw et al., 2011). The prefrontal changes observed in our SZ sample are congruent with the majority of the previous reports, which highlighted a consistent decrease in *alpha* band coherence in the frontal as well as parietal and occipital regions in SZ patients at rest, often associated with psychopathological indicators (for a review see Maran et al., 2016). In the study by Tauscher et al., *delta*, *theta* and *alpha* band coherences were observed to be lower in the prefrontal cortex in schizophrenia patients compared to healthy subjects (Tauscher et al., 1998). More recently, a reduction in the interhemispheric coherence of oscillations in the *theta* band in patients has been demonstrated (Koenig et al., 2001, Kam et al., 2013). However, early reports proposed an increase in coherence in low frequency bands: in the *delta* band (Wada et al., 1998), and in the *theta* band (Merrin et al., 1989).

In high frequency bands the coherence differences in SZ remain controversial. The decrease in coherence in SZ compared to controls in the *gamma* band has been reported in the past as the most conforming change in resting EEG in schizophrenia (Light et al., 2006; Bandyopadhyaya et al., 2011). In beta band EEG coherence, the noted differences in SZ patients are mixed. As Kam et al. showed, the inter-hemispheric *beta1* and *beta2* coherence was not significantly different between patients and healthy controls (Kam et al., 2013). In our previous studies, we found no differences in interhemispheric connections in high frequency *beta* band in the resting state in frontal areas (Gruzelier, 1999; Strelets et al., 2002). Our current result suggests the decreased coherence in the *beta1* band in SZ. Besides, it was inversely correlated with the severity of positive symptoms. The same result was obtained by Higashima et al. (2007) who demonstrated decreased *beta 1* coherence in frontal and temporal sites being associated with the severity of positive symptoms in both medicated and drug-naive schizophrenia patients in a longitudinal resting EEG study. The authors assumed that disconnection between the frontal and the temporal lobe in the left hemisphere might be associated with the generation of acute psychotic symptoms in schizophrenia. Supposedly, the discrepancies in reports on *beta* band coherence in SZ might be explained by inhomogeneity of the clinical groups and inclusion of the patients with various types of schizophrenia, various syndromes and trajectories of the disease. Since positive symptoms have a variety of presentations, the effects of particular symptoms e.g. the severity of delusions or hallucinations should be addressed in further studies.

Passing on to EEG during cognitive performance, the most general observation highlighted the brain regions that are activated during a task result in event-related desynchronization in the *alpha* band, whereas regions associated with task irrelevant and potentially interfering processes exhibit event-related synchronization (Klimesch, 2012). In our study, only healthy controls showed a local decrease in coherence (desynchronization) of the *alpha* band predominantly in the left frontal sites and parietal sites (symmetrically), i.e. in brain areas corresponding to mathematical operations. This supports the hypothesis that *alpha* band oscillations

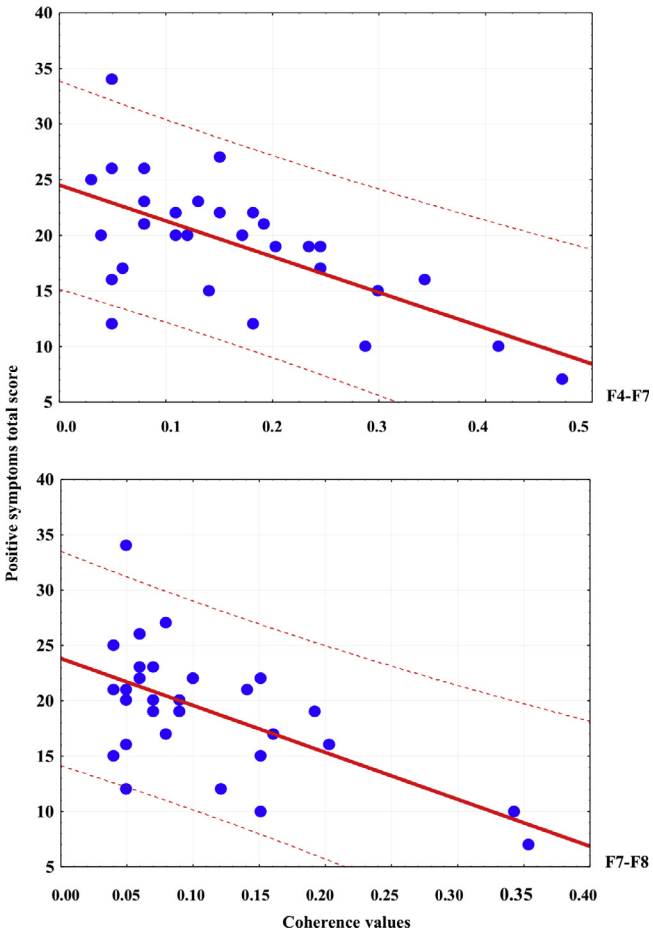

Fig. 4. Significant correlations in SZ group between inter-hemispheric coherence values in beta1 band in task condition and PANSS positive symptoms scores.

contribute to information processing (Klimesch, 2012). The decreased *alpha* coherence during the cognitive task is supported by previous studies (Hoffman et al., 1991; Morrison-Stewart et al., 1996). Abnormal *alpha* activity seems to be associated with working memory dysfunction (De Vico Fallani et al., 2010; Menon, 2010) and lower *alpha* band coherence in schizophrenia has been associated with psychomotor poverty (John et al., 2002). Interestingly, there was no selective decrease in *alpha* band synchronization in SZ patients, and the synchronization in patients of SA occurred in the sites that are not associated with mental arithmetic. One can assume that, if there is a desynchronization of the *alpha* band in the brain areas that are active during the task performance in the healthy population, then in SA the translocation of the oscillatory activity is driven by the mechanisms of synchronisation of *alpha* band activity in the cortical areas that are not usually activated during the task. This may indicate a deficiency in inhibitory mechanisms. In this regard, the inhibitory activity provided by *alpha* oscillations (having a role in the suppression of task-irrelevant neuronal processing as suggested by Palva and Palva (2011)) does not occur in SA. At the same time, the excessive decrease in spectral power and desynchronization in the *alpha* band in SZ presumably indicates the inordinate inhibitory mechanisms making the cognitive performance more demanding (Klimesch et al., 2007; Jensen and Mazaheri, 2010).

Given the potential functional roles in cognitive processing of other frequency bands, more specific and differentiated mechanisms can be affected as well. Recently, Tschentscher and Hauk (2016) suggested the functional role of the bands for an arithmetic problem-solving task, proposing the increased midline *theta power* for the retrieval strategy and reduced *alpha* and *beta* for procedural strategies, thus being differentially addressed in executive and retrieval demands. Furthermore, modulations of the *beta 1* band in frontal areas is well established and is associated with decision making (Pesaran et al., 2008; Wong et al., 2016) and relevant to selective attention processing (Spitzer and Blankenburg, 2011). As further proposed, the high frequency *beta band* has a similar functional role to *gamma* activity implying task-relevant engagement (Tallon-Baudry et al., 1998; Kornblith et al., 2016). In our SZ patient sample compared to the controls, we observed a decrease in coherence in the *beta1, beta2 and gamma2* bands in the right prefrontal and occipital, central and frontal areas. These findings, specifically the localization of these changes, are in line with Kissler et al., who showed reversed lateralization of the spectral power of *gamma* activity in the right frontal area during mental arithmetic task performance in SZ patients as opposed to the left frontal areas in healthy controls (Kissler et al., 2000), providing further support for a relation between abnormal *gamma* band oscillations and the disturbance of higher cognitive processes in schizophrenia. Other studies utilizing various cognitive tasks, though not requiring arithmetic abilities, showed similar results (for a review see Light et al., 2006; Uhlhaas et al., 2008; Uhlhaas and Singer, 2010, 2013). Spencer et al. observed decreased *gamma* band coherence in schizophrenic patients compared to healthy controls in response to visual sensory stimuli (Spencer et al., 2003). In another study by Bucci et al., gamma coherence was reduced in a subgroup of schizophrenia patients without predominant negative symptoms in response to an auditory oddball paradigm (Bucci et al., 2007).

On the contrary, SA patients showed a paradoxical increase in coherence in both low and high frequency bands (alpha, *beta1, beta2 and gamma* bands) during mental arithmetic task performance. We assume that such a non-selective increase in connectivity in the majority of the bands may be associated with the affective component in these patients. As a matter of fact, alterations in EEG coherence, notably in the *alpha* and *beta* bands, were demonstrated in affective disorders (Knott et al., 2001; Fingelkurts et al., 2007). Manic patients showed reduced *gamma* coherence in response to stimuli compared to healthy controls (Ozerdem et al., 2010) and the same results were observed in a group of bipolar patients (Özerdem et al., 2011). On the contrary, patients with major depressive disorders showed increased phase synchronization in the *theta, alpha and beta* bands primarily in the frontal-parietal and frontal temporal sites of the left hemisphere (Li et al., 2016). Since the majority of SA patients in the present study experienced a depressive type of SA, our results correspond well to the above-mentioned findings of Li et al (2016). This is also true with respect to the correlations with psychopathology, notably the correlations of task-related, left-lateralized coherence and positive symptoms.

Based on our previously reported findings for spectral power characteristics during rest and mental arithmetic task performance in the same samples (Garakh et al., 2015), we were able to identify the physiological patterns associated with each of the study groups. Thus, in healthy individuals, we found local changes in spectral power (SP) and coherence. Specifically, we detected an increase in SP in the frontal midline *theta* band, and SP and coherence in the anterior *beta1* and *gamma* bands, accompanied by a decrease in SP and coherence in the low frequency *alpha* band. In the SZ group, relatively to HC, we observed a decrease in SP in the *theta, alpha* and *beta 2* bands and the *alpha* coherence was significantly decreased. In the SA group, the characteristics of the SP during the mental arithmetic tasks mimicked the healthy group,

i.e. SP in the *gamma* band in the anterior sites of the left hemisphere increased, although the SP of the *alpha* band significantly decreased similar to the SZ group. In addition, similar to the SZ group, the SA group revealed a decrease in SP in the *theta* and *beta2* bands. At the same time, changes in coherence in the SA group are unique, being different from both the SZ and HC groups. Coherence in the anterior sites was elevated in both high and low frequency bands. Together, in healthy individuals, the increase and decrease of the EEG SP and synchronization across various frequency bands (each band has its own functional role) and in the particular locations can be considered as the optimal neural patterns needed for successful cognitive task performance. These mechanisms are violated in both psychotic disorders, but in distinct ways, being less inhibited in SA but at the same time demonstrating dysconnectivity in the frontal sites similar to SZ only in a resting state. Future studies should test our novel hypothesis that both these psychoses share aberrant functional connectivity but differ in the extent to which the inhibitory mechanisms are affected.

Several limitations should be mentioned. Firstly, the psychopathology was evaluated in the routine clinical examination and Structural Clinical Interview for PANSS, although the Structural Clinical and Diagnostic Interview for DSM-IV-TR axis I disorders (SKID) was not applied. This limitation was overcome by a number of qualified physicians involved in the diagnostic procedure and showing inter-rater reliability. Secondly, as we didn't focus specifically on factors influencing the coherence changes, we performed just an exploratory correlative analysis of coherence changes in patients' groups with the severity of psychopathological symptoms. The inverse association between beta 1 coherence and positive symptoms severity made us conclude that the decreased *beta 1* coherence in frontal sites during the task in schizophrenia might not originate only in the task performance as being rather predicted upon the severity of positive symptoms. Further investigation may provide stronger and more detailed evidence of this relation.

5. Conclusions

In summary, the synchronized activity of the neural networks is required for the successful information processing. Alterations in the behavioural task performance being at the first glance similar in the patients with schizophrenia and schizoaffective disorder, are presumably driven by distinct and even polar neural mechanisms. The distinctiveness and similarity indeed makes the scientific study of the schizophrenia spectrum disorders challenging (Heckers, 2009). The task related coherence was immensely diverse (and not the resting state coherence) and could better disentangle the clinical groups. The application of the mental arithmetic task that is sensitive to the frontal activity being a hallmark of schizophrenia and schizophrenia spectrum disorders allowed to elicit the commonalities in coherences and differences in EEG bands reflecting predominantly inhibitory processes required for effective information processing. This study provides an evidence of neurophysiological distinction of these two disorders from the schizophrenia spectrum.

Acknowledgments

We thank Dr. A. Dorodnova who helped with the enrolment of patients from the Early Intervention Centre, Moscow Research Institute of Psychiatry, and Dr C. Hampson, who assisted with proofreading of the manuscript. The project was supported by European Regional Development Fund, by the project "Sustainability for the National Institute of Mental Health" (grant number LO1611) and Czech Research Council (grant number AZV 17-32957A) and supported by the Russian Academy of Sciences.

Conflicts of interest

The authors have declared no competing interests.

References

Bandyopadhyaya D, Nizamie SH, Pradhan N, Bandyopadhyaya A. Spontaneous gamma coherence as a possible trait marker of schizophrenia: an explorative study. Asian J Psychiatr 2011;4:172–7. https://doi.org/10.1016/j.ajp.2011.06.006.

Bendat JS, Piesol AG. Random Data: analysis and measurement procedures. 4th ed. Wiley; 2010.

Bowyer SM. Coherence a measure of the brain networks: past and present. Neuropsychiatr Electrophysiol 2016;2. https://doi.org/10.1186/s40810-015-0015-7.

Boutros NN, Arfken C, Galderisi S, Warrick J, Pratt G, Iacono W. The status of spectral EEG abnormality as a diagnostic test for schizophrenia. Schizophr Res 2008;99:225–37. https://doi.org/10.1016/j.schres.2007.11.020.

Brenner CA, Krishnan GP, Vohs JL, Ahn WY, Hetrick WP, Morzorati SL, et al. Steady state responses: electrophysiological assessment of sensory function in schizophrenia. Schizophr Bull 2009;35:1065–77. https://doi.org/10.1093/schbul/sbp091.

Bucci P, Mucci A, Merlotti E, Volpe U, Galderisi S. Induced gamma activity and event-related coherence in schizophrenia. Clin EEG Neurosci 2007;38:96–104. https://doi.org/10.1177/155005940703800212.

Decker SL, Fillmore PT, Roberts A. Coherence: the measurement and application of brain connectivity. NeuroRegulation 2017;4:3–13. https://doi.org/10.15540/nr.4.1.3.

DeStefano D, LeFevre JA. The role of working memory in mental arithmetic. Eur J Cogn Psychol 2004;16:353–86. https://doi.org/10.1080/09541440244000328.

De Vico Fallani F, Maglione A, Babiloni F, Mattia D, Astolfi L, Vecchiato G, et al. Cortical network analysis in patients affected by schizophrenia. Brain Topogr 2010;23:214–20. https://doi.org/10.1007/s10548-010-0133-2.

Di Lorenzo G, Daverio A, Ferrentino F, Santarnecchi E, Ciabattini F, Monaco L, et al. Altered resting-state EEG source functional connectivity in schizophrenia: the effect of illness duration. Front Hum Neurosci 2015;9:234. https://doi.org/10.3389/fnhum.2015.00234.

Douw L, Schoonheim MM, Landi D, van der Meer ML, Geurts JJ, Reijneveld JC, et al. Cognition is related to resting-state small-world network topology: a magnetoencephalographic study. Neuroscience 2011;175:169–77. https://doi.org/10.1016/j.neuroscience.2010.11.039.

Du Y, Pearlson GD, Lin D, Sui J, Chen J, Salman M, et al. Identifying dynamic functional connectivity biomarkers using GIG-ICA: application to schizophrenia, schizoaffective disorder, and psychotic bipolar disorder. Hum Brain Mapp 2017;38:2683–708. https://doi.org/10.1002/hbm.23553.

Fingelkurts AA, Fingelkurts AA, Rytsälä H, Suominen K, Isometsä E, Kähkönen S. Impaired functional connectivity at EEG alpha and theta frequency bands in major depression. Hum Brain Mapp 2007;28:247–61. https://doi.org/10.1002/hbm.20275.

Fries P. Rhythms for cognition: communication through coherence. Neuron 2015;88:220–35. https://doi.org/10.1016/j.neuron.2015.09.034.

Fries P. A mechanism for cognitive dynamics: neuronal communication through neuronal coherence. Trends Cogn Sci 2005;9:474–80. https://doi.org/10.1016/j.tics.2005.08.011.

Friston KJ. Schizophrenia and the disconnection hypothesis. Acta Psychiatr Scand Suppl 1999;395:68–79. https://doi.org/10.1111/j.1600-0447.1999.tb05985.x.

Garakh Z, Zaytseva Y, Kapranova A, Fiala O, Horacek J, Shmukler A, et al. EEG correlates of a mental arithmetic task in patients with first episode schizophrenia and schizoaffective disorder. Clin Neurophysiol 2015;126:2090–8. https://doi.org/10.1016/j.clinph.2014.12.031.

Greicius M. Resting-state functional connectivity in neuropsychiatric disorders. Curr Opin Neurol 2008;21:424–30. https://doi.org/10.1097/WCO.0b013e328306f2c5.

Gruzelier JH. Functional neuropsychophysiological asymmetry in schizophrenia: a review and reorientation. Schizophr Bull 1999;25:91–120. https://doi.org/10.1093/oxfordjournals.schbul.a033370.

Heckers S. Is schizoaffective disorder a useful diagnosis. Curr Psychiatry Rep 2009;11:332–7.

Heinrichs RW, Zakzanis KK. Neurocognitive deficit in schizophrenia: a quantitative review of the evidence. Neuropsychology 1998;12:426–45. https://doi.org/10.1037/0894-4105.12.3.426.

Higashima M, Takeda T, Kikuchi M, Nagasawa T, Hirao N, Oka T, et al. State-dependent changes in intrahemispheric EEG coherence in patients with acute exacerbation of schizophrenia. Psychiatr Res 2007;149:41–7.

Hillebrand A, Barnes GR, Bosboom JL, Berendse HW, Stam CJ. Frequency-dependent functional connectivity within resting state networks: an atlas-based MEG beamformer solution. Neuroimage 2012;59:3909–21. https://doi.org/10.1016/j.neuroimage.2011.11.005.

Hoffman RE, Buchsbaum MS, Escobar MD, Makuch RW, Nuechterlein KH, Guich SM. EEG coherence of prefrontal areas in normal and schizophrenic males during perceptual activation. J Neuropsychiatry Clin Neurosci 1991;3:169–75. https://doi.org/10.1176/jnp.3.2.169.

Jensen O, Mazaheri A. Shaping functional architecture by oscillatory alpha activity: gating by inhibition. Front Hum Neurosci 2010;4:186. https://doi.org/10.3389/fnhum.2010.00186.
John JP, Khanna S, Pradhan N, Mukundan CR. EEG alpha coherence and psychopathological dimensions of schizophrenia. Indian J Psychiatry 2002;44:97–107.
Kam JW, Bolbecker AR, O'Donnell BF, Hetrick WP, Brenner CA. Resting state EEG power and coherence abnormalities in bipolar disorder and schizophrenia. J Psychiatr Res 2013;47:1893–901. https://doi.org/10.1016/j.jpsychires.2013.09.009.
Keefe RSE. Should cognitive impairment be included in the diagnostic criteria for schizophrenia? World Psychiatry 2008;7:22–8. https://doi.org/10.1002/j.2051-5545.2008.tb00142.x.
Kissler J, Müller MM, Fehr T, Rockstroh B, Elbert T. MEG gamma band activity in schizophrenia patients and healthy subjects in a mental arithmetic task and at rest. Clin Neurophysiol 2000;111:2079–87. https://doi.org/10.1016/S1388-2457(00)00425-9.
Klimesch W, Sauseng P, Hanslmayr S. EEG alpha oscillations: the inhibition-timing hypothesis. Brain Res Rev 2007;53:63–88. https://doi.org/10.1016/j.brainresrev.2006.06.003.
Klimesch W. Alpha-band oscillations, attention, and controlled access to stored information. Trends Cogn Sci 2012;16:606–17. https://doi.org/10.1016/j.tics.2012.10.007.
Knott V, Mahoney C, Kennedy S, Evans K. EEG power, frequency, asymmetry and coherence in male depression. Psychiatry Res 2001;106:123–40. https://doi.org/10.1016/S0925-4927(00)00080-9.
Koenig T, Lehmann D, Saito N, Kuginuki T, Kinoshita T, Koukkou M. Decreased functional connectivity of EEG theta-frequency activity in first-episode, neuroleptic-naïve patients with schizophrenia: preliminary results. Schizophr Res 2001;50:55–60. https://doi.org/10.1016/S0920-9964(00)00154-7.
Kornblith S, Buschman TJ, Miller EK. Stimulus load and oscillatory activity in higher cortex. Cereb Cortex 2016;26:3772–84. https://doi.org/10.1093/cercor/bhv182.
Landin-Romero R, Canales-Rodríguez EJ, Kumfor F, Moreno-Alcazár A, Madre M, Maristany T, et al. Surface-based brain morphometry and diffusion tensor imaging in schizoaffective disorder. Aust N Z J Psychiatry 2017;51:42–54. https://doi.org/10.1177/0004867416631827.
Lewis DA. Cortical circuit dysfunction and cognitive deficits in schizophrenia–implications for preemptive interventions. Eur J Neurosci 2012;35:1871–8. https://doi.org/10.1111/j.1460-9568.2012.08156.x.
Li Y, Kang C, Qu X, Zhou Y, Wang W, Hu Y. Depression-related brain connectivity analyzed by EEG event-related phase synchrony measure. Front Hum Neurosci 2016;10:477. https://doi.org/10.3389/fnhum.2016.00477.
Light GA, Hsu JL, Hsieh MH, Meyer-Gomes K, Sprock J, Swerdlow NR, et al. Gamma band oscillations reveal neural network cortical coherence dysfunction in schizophrenia patients. Biol Psychiatry 2006;60:1231–40. https://doi.org/10.1016/j.biopsych.2006.03.055.
Madre M, Canales-Rodríguez EJ, Ortiz-Gil J, Murru A, Torrent C, Bramon E, et al. Neuropsychological and neuroimaging underpinnings of schizoaffective disorder: a systematic review. Acta Psychiatr Scand 2016;134:16–30. https://doi.org/10.1111/acps.12564.
Maran M, Grent-'t-ong T, Uhlhaas PJ. Electrophysiological insights into connectivity anomalies in schizophrenia: a systematic review. Neuropsychiatr Electrophysiol 2016;2:6. https://doi.org/10.1186/s40810-016-0020-5.
Menon V. Developmental cognitive neuroscience of arithmetic: implications for learning and education. ZDM 2010;42:515–25. https://doi.org/10.1007/s11858-010-0242-0.
Merrin EL, Floyd TC, Fein G. EEG coherence in unmedicated schizophrenic patients. Biol Psychiatry 1989;25:60–6. https://doi.org/10.1016/0006-3223(89)90147-9.
Morrison-Stewart SL, Williamson PC, Corning WC, Kutcher SP, Merskey H. Coherence on electroencephalography and aberrant functional organisation of the brain in schizophrenic patients during activation tasks. Br J Psychiatry 1991;159:636–44. https://doi.org/10.1192/bjp.159.5.636.
Morrison-Stewart SL, Velikonja D, Corning WC, Williamson P. Aberrant interhemispheric alpha coherence on electroencephalography in schizophrenic patients during activation tasks. Psychol Med 1996;26:605–12. https://doi.org/10.1017/S0033291700035674.
Novototsky-Vlasov V, Garah JV, Kovalev VP. A method for repetitive artifact suppression in multichannel EEG recordings. Hum Physiol 2007;33:231–5. https://doi.org/10.1134/S0362119707020156.
Ozerdem A, Güntekin B, Saatçi E, Tunca Z, Başar E. Disturbance in long distance gamma coherence in bipolar disorder. Prog Neuropsychopharmacol Biol Psychiatry 2010;34:861–5. https://doi.org/10.1016/j.pnpbp.2010.04.001.
Özerdem A, Güntekin B, Atagün I, Turp B, Başar E. Reduced long distance gamma (28–48 Hz) coherence in euthymic patients with bipolar disorder. J Affect Disorders 2011;132:325–32. https://doi.org/10.1016/j.jad.2011.02.028.
Palva S, Palva JM. Functional roles of alpha-band phase synchronization in local and large-scale cortical networks. Front Psychol 2011;2:204. https://doi.org/10.3389/fpsyg.2011.00204.
Pesaran B, Nelson MJ, Andersen RA. Free choice activates a decision circuit between frontal and parietal cortex. Nature 2008;453:406–9. https://doi.org/10.1038/nature06849.
Popov TG, Rockstroh BS, Popova P, Carolus AM, Miller GA. Dynamics of alpha oscillations elucidate facial affect recognition in schizophrenia. Cogn Affect Behav Neurosci 2014;14:364–77. https://doi.org/10.3758/s13415-013-0194-2.
Potkin SG, Ford JM. Widespread cortical dysfunction in schizophrenia: the FBIRN imaging consortium. Schizophr Bull 2009;35:15–8. https://doi.org/10.1093/schbul/sbn159.
Schultz CC, Fusar-Poli P, Wagner G, Koch K, Schachtzabel C, Gruber O, et al. Multimodal functional and structural imaging investigations in psychosis research. Eur Arch Psychiatry Clin Neurosci 2012;262:S97–S106. https://doi.org/10.1007/s00406-012-0360-5.
Shmukler AB, Gurovich IY, Agius M, Zaytseva Y. Long-term trajectories of cognitive deficits in schizophrenia: a critical overview. Eur Psychiatry 2015;30:1002–10. https://doi.org/10.1016/j.eurpsy.2015.08.005.
Spencer KM, Nestor PG, Niznikiewicz MA, Salisbury DF, Shenton ME, McCarley RW. Abnormal neural synchrony in schizophrenia. J Neurosci 2003;23:7407–11.
Spitzer B, Blankenburg F. Stimulus-dependent EEG activity reflects internal updating of tactile working memory in humans. Proc Natl Acad Sci U S A 2011;108:8444–9. https://doi.org/10.1073/pnas.1104189108.
Stephan KE, Friston KJ, Frith CD. Dysconnection in schizophrenia: from abnormal synaptic plasticity to failures of self-monitoring. Schizophr Bull 2009;35:509–27. https://doi.org/10.5167/uzh-19544.
Strelets VB, Novototsky-Vlasov VY, Golikova JV. Cortical connectivity in high frequency beta-rhythm in schizophrenics with positive and negative symptoms. Int J Psyhophysiol 2002;44:101–15. https://doi.org/10.1016/S0167-8760(01)00196-9.
Tallon-Baudry C, Bertrand O, Peronnet F, Pernier J. Induced gamma-band activity during the delay of a visual short-term memory task in humans. J Neurosci 1998;18:4244–54.
Tauscher J, Fischer P, Neumeister A, Rappelsberger P, Kasper S. Low frontal electroencephalographic coherence in neuroleptic-free schizophrenic patients. Biol Psychiatry 1998;44:438–47. https://doi.org/10.1016/S0006-3223(97)00428-9.
Tschentscher N, Hauk O. Frontal and parietal cortices show different spatiotemporal dynamics across problem-solving stages. J Cogn Neurosci 2016;28:1098–110. https://doi.org/10.1162/jocn_a_00960.
Uhlhaas PJ, Haenschel C, Nikolić D, Singer W. The role of oscillations and synchrony in cortical networks and their putative relevance for the pathophysiology of schizophrenia. Schizophr Bull 2008;34:927–43. https://doi.org/10.1093/schbul/sbn062.
Uhlhaas PJ, Singer W. Abnormal neural oscillations and synchrony in schizophrenia. Nat Rev Neurosci 2010;11:100–13. https://doi.org/10.1038/nrn2774.
Uhlhaas PJ, Singer W. High-frequency oscillations and the neurobiology of schizophrenia. Dialogues Clin Neurosci 2013;15:301–13.
van Haren NE, Bakker SC, Kahn RS. Genes and structural brain imaging in schizophrenia. Curr Opin Psychiatry 2008;21:161–7. https://doi.org/10.1097/YCO.0b013e3282f4f25b.
Winterer G, Egan MF, Rädler T, Hyde T, Coppola R, Weinberger DR. An association between reduced interhemispheric EEG coherence in the temporal lobe and genetic risk for schizophrenia. Schizophr Res 2001;49:129–43. https://doi.org/10.1016/S0920-9964(00)00128-6.
Wong YT, Fabiszak MM, Novikov Y, Daw ND, Pesaran B. Coherent neuronal ensembles are rapidly recruited when making a look-reach decision. Nat Neurosci 2016;19:327–34. https://doi.org/10.1038/nn.4210.
Wada Y, Nanbu Y, Kikuchi M, Koshino Y, Hashimoto T. Aberrant functional organization in schizophrenia: analysis of EEG coherence during rest and photic stimulation in drug-naive patients. Neuropsychobiology 1998;38:63–9. https://doi.org/10.1159/000026518.